

SENTHILKUMARAN P

SOFTWARE ENGINEER — BACKEND & DISTRIBUTED SYSTEMS

+91 94437 26090 | senthil2005kumaran@gmail.com | Salem, Tamil Nadu

[linkedin.com/in/senthilkumaran75](https://www.linkedin.com/in/senthilkumaran75) | github.com/SenthilkumaranPSK | senthilkumaran.vercel.app

PROFESSIONAL SUMMARY

Backend Engineer whose software runs in production for paying clients, not in demos. Builds for correctness under real conditions - currency stored as integer paise with GST round-off reconciliation, append-only audit logging, 18 schema migrations with zero data loss - and tests accordingly, with property-based (Hypothesis) and AWS-mocked (moto) suites across unit, integration and performance layers. Works across Python and TypeScript: FastAPI and Fastify services, PostgreSQL and Prisma data layers, S3-backed model registries with versioned metadata. Recent 2026 graduate.

SKILLS

Languages	Python, TypeScript, SQL
Backend	FastAPI, Uvicorn, Pydantic v2, Fastify, Express, REST API design, API key authentication, per-client rate limiting
Data	PostgreSQL, SQLAlchemy 2.0, Prisma, SQLite (WAL), Amazon S3 artifact storage
Testing	pytest, pytest-cov, pytest-asyncio, Hypothesis (property-based), moto (AWS mocking), ruff, mypy
Infrastructure	Docker, GitHub Actions CI, boto3/AWS, Prometheus, structured JSON logging, Linux
ML Systems	Model registry and versioning, Prophet, SARIMAX, statsmodels, scikit-learn

WORK EXPERIENCE

Web & Software Developer

Freelance | Salem, Tamil Nadu | Jun 2026 – Present

github.com/SenthilkumaranPSK/Maestro-billing

- Built and shipped a GST-compliant billing system now in daily production for a paying client, using React 18, Vite, Tailwind, Fastify, Prisma and SQLite, with PDF invoices, 80mm thermal receipts and automatic WhatsApp delivery.
- Implemented integer-based currency handling in paise with CGST/SGST whole-rupee reconciliation, preventing floating-point calculation errors.
- Designed an append-only audit log that excludes cancelled bills from revenue, and maintained the schema across 18 Prisma migrations with zero data loss and automated daily backups.
- Built and deployed a company website for an IT services client.

AI Developer Intern

Python and Ladder (Uraikathai) | Salem, Tamil Nadu | Jan 2026 – Jun 2026

- Built AI-assisted image and video generation pipelines using SDXL, img2img and ControlNet, running latent-diffusion workflows in ComfyUI on RunPod cloud GPUs.
- Contributed frontend development to the interactive fiction platform alongside the generative research work.
- Researched character consistency and scene coherence across generated sequences through prompt orchestration and parameter experimentation under a human-in-the-loop evaluation framework.

AI & ML Intern

Apex Seekers Edtech Pvt Ltd | Salem, Tamil Nadu | Jul 2025 – Aug 2025

- Built end-to-end machine learning pipelines in Python, pandas and scikit-learn covering preprocessing, feature engineering, model training and hyperparameter search.
- Ran comparative model evaluation across multiple algorithm families to select production candidates against task-appropriate metrics.

Data Analyst Intern

Stack Queue | Salem, Tamil Nadu | Aug 2024 – Sep 2024

- Profiled a 376-resort European ski dataset across 18 infrastructure and pricing features, quantifying distribution skew to flag outlier-driven variables ahead of modelling.
- Built an automated column-typing and distribution-analysis routine covering all 13 numeric fields with histograms, KDE plots and boxplots.

PROJECTS

Demand Forecasting Platform — Model Registry & Forecast API

Python · FastAPI · Pydantic v2 · PostgreSQL · SQLAlchemy 2.0 · boto3/S3 · Amazon Forecast · Prophet · statsmodels
[github.com/SenthilkumaranPSK/Sales-Demand-Predictor](#)

- Built a model registry storing artifacts in Amazon S3 with versioned metadata in PostgreSQL, supporting multi-model comparison per product.
- Served forecasts with 50/80/90% confidence intervals over a FastAPI service, secured with key-based authentication and per-client rate limiting.
- Added health endpoints and structured JSON logging for operational visibility.
- Tested with pytest, pytest-asyncio, Hypothesis property-based testing and moto AWS mocking across unit, integration and performance suites.

Nuclei Instance Segmentation — Classical CV vs U-Net

Python · PyTorch · torchvision · OpenCV · scikit-image · FastAPI · pytest
[github.com/SenthilkumaranPSK/Biomedical-Image-Analysis](#)

- Benchmarked a classical OpenCV pipeline against a U-Net on BBBC039 (Broad Institute, 200 fluorescence microscopy images, official 100/50/50 split).
- Separated touching nuclei using distance-transform seeded watershed and a learned boundary channel, evaluating on Dice, IoU and instance-level F1 rather than pixel accuracy, which is misleading under class imbalance.
- Applied percentile normalisation over min-max scaling to preserve dim nuclei across 16-bit dynamic range; grid-searched parameters on train and reported on validation.
- Served inference through a FastAPI endpoint backing a live demo.

RAG Knowledge Assistant

Python · sentence-transformers · Chroma · BM25 · cross-encoder reranking · pdfplumber
[github.com/SenthilkumaranPSK/RAG-Chatbot](#)

- Built a hybrid retrieval pipeline blending dense vector search with BM25 keyword matching, followed by cross-encoder reranking.
- Added query rewriting and sentence-aware overlapping chunking to improve recall across long documents.
- Grounded every answer in cited sources; runs fully locally except the generation call.

AUREXIS AI — Multi-Agent Personal Finance System

Python · FastAPI · React/Vite · Ollama (deepseek-v2) · scikit-learn · Docker · GitHub Actions
[github.com/SenthilkumaranPSK/AUREXIS_AI](#) · [aurexis-ai.vercel.app](#)

- Architected a FastAPI service coordinating specialised agents for portfolio analysis, transaction monitoring and recommendations.
- Implemented weighted risk scoring and time-series forecasting to evaluate financial stability, plus a scenario simulation module for testing hypothetical decisions.
- Containerised with Docker and Docker Compose behind a GitHub Actions CI pipeline; ran entirely on local inference, removing per-token API cost.
- Published as first author of seven: AUREXIS AI: A Cognitive AI-based System for Real-Time Financial Analysis and Decision Support, IJIRT, Vol. 12, Issue 12, pp.3521-3529, May 2026. [ijirt.org/article?manuscript=200876](#)

EDUCATION

B.Tech, Artificial Intelligence & Data Science

Dhirajlal Gandhi College of Technology | Salem, Tamil Nadu | May 2026
CGPA 7.85 / 10

ACHIEVEMENTS

- Google Agentic AI Day 2025 Hackathon — shortlisted at national level.
- AI for Bharat — Certificate of Recognition for a GenAI content summariser built on Amazon Bedrock (AWS × Hack2Skill).
- Salem District-Level Forest Fire Prevention Hackathon — 2nd place, Tamil Nadu State Planning Commission.

CERTIFICATIONS

- NPTEL Design Thinking: A Primer — IIT Madras (Elite, 2026).
- Google Cloud Skill Badge — Inspect Rich Documents with Gemini Multimodality and Multimodal RAG.
- Course on Machine Learning — LIVEWIRE (Sep 2024 – Mar 2025).

LANGUAGES

Tamil · English